

# Metaheuristic Hyperparameter Optimization for Intraday Financial Trend Classification: A Benchmark Across Classifier Backbones

## Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound  $0.60 \text{ MCC} + 0.40 F_1$  fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

**Keywords:** Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

## 1 Introduction

Futures contracts are central instruments for hedging and speculation because their prices respond rapidly to expectations, order flow, and changing risk conditions. For computational prediction, this responsiveness is attractive but difficult: five-minute price series contain short-lived patterns, regime changes, and substantial noise. The Brazilian Mini-Index futures contract (WIN), referenced to the Ibovespa index, is a relevant setting because it is highly liquid and supports short-horizon intraday trading.

Public BovDB resources make this setting reproducible for financial machine-learning research [1].

The blinded precursor to the present study investigated the same problem through technical indicators, Information Gain feature selection, and genetic-algorithm (GA) tuning of Random Forest, Support Vector Machine, and Multilayer Perceptron classifiers [2]. Its central premise remains useful: compact technical representations and global hyperparameter search can improve nonlinear classifiers in noisy financial data. Yet its experimental design was intentionally exploratory. It relied on randomized cross-validation, compared only one metaheuristic, did not equalize search budgets across optimizer families, and stopped at predictive metrics. Those limitations prevent a reliable claim about either temporal generalization or the comparative merit of an optimizer.

This study retains the precursor’s practical narrative—technical indicators provide a compact representation of intraday price dynamics, and metaheuristics search the mixed discrete–continuous hyperparameter space more broadly than manual tuning—while changing the experimental question. Rather than asking whether GA-tuned models exceed an untuned baseline, we ask how a fixed optimization budget and the choice of validation fitness affect the behaviour of multiple model–optimizer combinations on a locked chronological test block.

Three design issues motivate this question. First, random shuffling in financial time series can allow future observations to influence model selection. Second, metaheuristics can appear superior simply because they receive more candidate evaluations than a comparator. Third, accuracy is not interchangeable with an imbalance-robust criterion such as Matthews correlation coefficient (MCC), nor does either predictive metric by itself establish economic usefulness [3]. The distinction between the score used to select a candidate and the score reported after testing is especially important in financial classification.

The present work makes six contributions:

- **Reframe** the GA-based precursor as an equal-budget comparison of Random Search (RS), GA, Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimization (GWO).
- **Replace** randomized validation with a chronological 60/20/20 train/validation/test protocol on 15,057 five-minute WIN bars.
- **Reuse** the compact seven-indicator Information-Gain representation inherited from the blinded precursor, thereby holding the input representation fixed while comparing optimizers.
- **Separate** two validation objectives: compound  $0.60 \text{ MCC} + 0.40 F_1$  fitness and conventional validation accuracy.
- **Report** MCC, accuracy, and  $F_1$  only on the locked test set after candidate selection, with direct fitness comparisons limited to paired seeds that are available in both protocols.
- **Extend** predictive assessment with exploratory within-protocol non-parametric tests and an intraday economic evaluation based on total profit and drawdown.

The article is organized as follows. Section 2 situates the work in the literature and clarifies its relationship with the precursor. Section 3 describes the shared search controls and the two fitness protocols. Section 4 reports predictive, statistical, and economic results. Sections 5 and 6 discuss the implications and limitations.

## 2 Related Work and Research Positioning

Technical indicators remain a common bridge between market data and supervised learning. Moving averages, exponential smoothing, and volatility transformations summarize local momentum and dispersion; however, a large indicator pool can produce redundant inputs and unstable models. Information-theoretic filters such as Information Gain offer a computationally simple way to retain attributes that reduce uncertainty about the target class [4, 5]. The blinded precursor study adopted this strategy to reduce 66 candidate descriptors to seven indicators and then used GA to tune classifier hyperparameters [2].

Hyperparameter tuning is a natural target for metaheuristics because neural and kernel models combine continuous variables, integer architecture choices, and categorical settings. GA and DE introduce variation through recombination and mutation; PSO and GWO use population-based attraction mechanisms; Random Search supplies an unguided but essential budget-matched baseline [6–10]. Their comparative behaviour is difficult to interpret unless every optimizer evaluates the same number of candidate configurations.

Recent financial-learning work has also strengthened the case for temporally ordered assessment. When observations are shuffled, a model can learn distributional information from a later market state while being evaluated on an earlier state. A sequential holdout avoids that direct leakage, although it does not eliminate the broader challenge of market-regime shift. Similarly, MCC offers a balanced summary of binary predictions and should be kept distinct from conventional accuracy when the selection objective is discussed [3].

Table 1 positions the article as a substantial methodological extension, rather than a repetition, of the precursor. The change is not merely an increase in the number of models: it changes the validation logic, comparator set, scope of evidence, and economic endpoint.

The resulting research position is deliberately conservative. This manuscript does not claim that one fitness is globally superior from the currently available three common seeds. It instead establishes a controlled comparison and makes clear which analyses describe the fitness contrast and which test model separation within a fixed protocol.

## 3 Materials and Methods

This section presents the methodology adopted in this study. Figure 1 provides an overview of the flow from data acquisition to final model evaluation. It preserves the practical sequence of the IJCNN precursor—input data, technical indicators, feature selection, optimization, model fitting, and evaluation—while inserting the temporal controls and the two validation objectives required by the expanded benchmark.

**Table 1** Methodological progression from the GA-based precursor to the present benchmark.

| Component            | Precursor study                   | Present study                                           |
|----------------------|-----------------------------------|---------------------------------------------------------|
| Validation           | Randomized cross-validation       | Chronological 60/20/20 holdout                          |
| Optimizer comparison | GA only                           | RS, GA, PSO, DE, and GWO                                |
| Search budget        | Generation-based GA configuration | 1,000 candidate evaluations per optimizer and seed      |
| Selection objective  | Cross-validation accuracy         | MCC/ $F_1$ fitness or validation accuracy               |
| Evaluation           | Predictive metrics                | Locked predictive, statistical, and economic assessment |

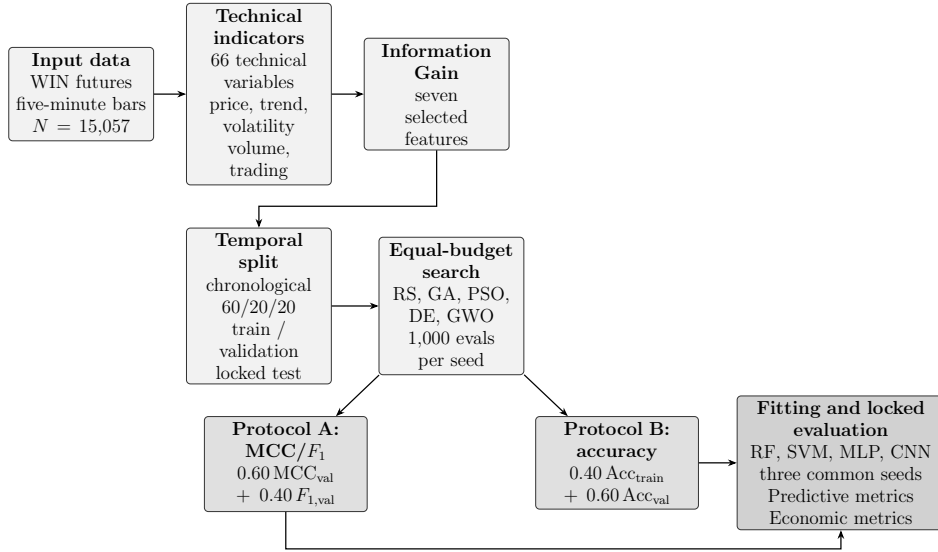
**Fig. 1** Overview of the proposed methodology. The benchmark reuses the precursor’s fixed Information-Gain representation while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

Table 2 consolidates the decisions that define the experimental unit. It is included to make clear which conditions are held fixed while optimizers and classifier backbones vary. Detailed search domains and algorithm-specific update settings are provided in the subsequent subsections.

### 3.1 Input Data

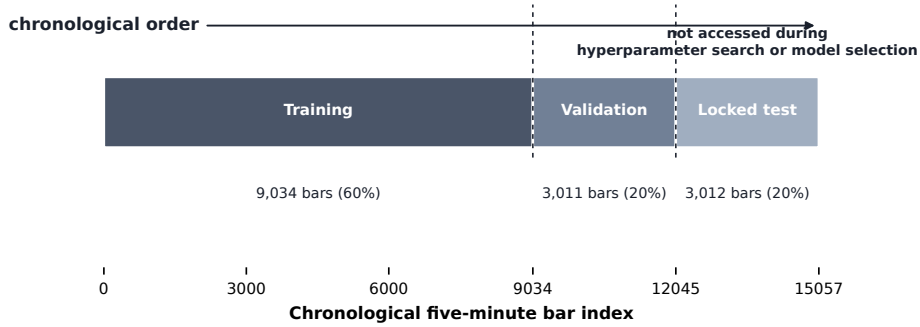
The data were obtained from the BovDB resources [1]. The analyzed subset contains 15,057 clean five-minute records of the WIN contract from January to June 2024. WIN is referenced to the Ibovespa index and was selected because its liquidity and frequent

**Table 2** Reproducible experimental protocol. All candidates within a model–optimizer–seed cell use the same fixed representation, chronological partition, and evaluation budget.

| Component               | Specification                                                                                                                                            |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data                    | 15,057 five-minute WIN records from January–June 2024; next-bar binary direction target.                                                                 |
| Input representation    | Fixed seven-indicator Information-Gain mask inherited from the blinded precursor.                                                                        |
| Partition               | Chronological 60/20/20 split: 9,034 training, 3,011 validation, and 3,012 locked-test observations.                                                      |
| Backbones               | Random Forest, Support Vector Machine, Multilayer Perceptron, and 1D Convolutional Neural Network.                                                       |
| Search methods          | Random Search, Genetic Algorithm, Particle Swarm Optimization, Differential Evolution, and Grey Wolf Optimizer.                                          |
| Budget and repetition   | 1,000 candidate evaluations per optimizer, model, and seed; common seeds 1–3. Population methods use 10 candidates for 100 iterations.                   |
| Selection objectives    | Protocol A: $0.60 \text{MCC}_{\text{val}} + 0.40 F_{1,\text{val}}$ ; Protocol B: $0.40 \text{Acc}_{\text{train}} + 0.60 \text{Acc}_{\text{val}}$ .       |
| Preprocessing and refit | Standardizer fitted on training only during search; final selected configuration refitted on training plus validation before one locked-test evaluation. |
| Reported endpoints      | Locked-test accuracy, MCC, $F_1$ , exploratory model-level inference, and fixed-rule intrabar economic outcomes.                                         |

intraday movement make it suitable for short-horizon computational prediction. The binary target indicates the next-bar direction: 0 for downtrend and 1 for uptrend.

The records remain in chronological order. The first 9,034 observations are used for training, the subsequent 3,011 observations for validation, and the final 3,012 observations for locked testing. No random shuffling is applied. Figure 2 makes this separation explicit; the test observations are unavailable during hyperparameter selection and final model choice.



**Fig. 2** Chronological data partition used in every experiment. Training, validation, and locked test segments occupy consecutive periods of the WIN five-minute series.

### 3.2 Generation of Technical Analysis Indicators

Technical indicators were derived from open, high, low, close, trade volume, and transaction-frequency information. Simple Moving Average (SMA), Exponential Moving Average (EMA), and dispersion measures were computed over short windows (3, 5, 7, and 9 periods) to retain sensitivity to intraday changes. For example, the difference between the seven- and three-period SMA is

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3. \quad (1)$$

Together, the construction procedure produces 66 descriptors, including moving-average differences and normalized counterparts. For each optimizer evaluation, feature standardization is fitted on the training partition only and then applied unchanged to the validation partition. For the final locked-test evaluation, a new standardization transform is fitted on the combined training-validation block and applied to the test block only.

### 3.3 Feature Selection Method

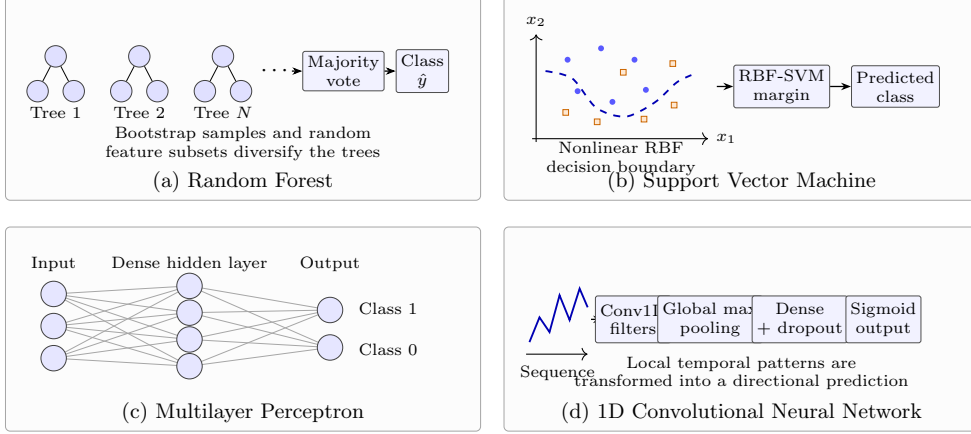
The current benchmark reuses a fixed seven-variable mask obtained by Information Gain in the blinded precursor rather than re-estimating the filter in each current split [2]. The fixed mask contains  $\text{EMA}_{5-3}$ , normalized  $\text{EMA}_{5-3}$ ,  $\text{EMA}_{7-3}$ , normalized  $\text{EMA}_{7-3}$ , normalized  $\text{SMA}_9$ ,  $\text{EMA}_{9-3}$ , and normalized  $\text{EMA}_7$ . It is therefore applied identically to every backbone, optimizer, and seed, isolating the optimizer comparison from changes in input representation. This choice does not independently revalidate the feature-selection stage on the present training segment; the reported evidence concerns hyperparameter search conditional on this inherited representation.

### 3.4 Implemented Classifier Architectures and Search Spaces

To benchmark metaheuristic search performance across fundamentally distinct hypothesis spaces, four classifier backbones were implemented: Random Forest (RF), Support Vector Machines (SVM), Multilayer Perceptron (MLP), and 1D Convolutional Neural Networks (1D-CNN). Figure 3 illustrates the structural decision mechanisms and layer topologies of these four implemented architectures.

The hyperparameter search space bounds  $\theta \in \mathcal{S}$  for each implemented model are defined as follows:

- **RF**: Number of trees  $n_{\text{estimators}} \in [50, 500]$ , maximum tree depth  $d_{\text{max}} \in [5, 50]$ , minimum split samples  $s_{\text{split}} \in [2, 20]$ , minimum leaf samples  $s_{\text{leaf}} \in [1, 20]$ , and categorical choices for `max_features` (`{sqrt, log2, None}`) and `class_weight` (`{None, balanced}`).
- **SVM**: Soft-margin penalty  $C \in [10^{-3}, 10^3]$  and kernel bandwidth  $\gamma \in [10^{-5}, 10^1]$ , both sampled on logarithmic scales, with linear and radial-basis-function kernels considered.
- **MLP**: Hidden-layer width  $n_{\text{hidden}} \in [5, 500]$ , L2 penalty coefficient  $\alpha \in [10^{-8}, 10^{-2}]$ , initial learning rate  $\eta_0 \in [10^{-8}, 10^{-2}]$ , dropout  $p_{\text{drop}} \in [0, 0.10]$ , and batch size in  $\{16, 32, 64, 80, 112, 128\}$ .



**Fig. 3** Simplified vector schematics of the four classifier backbones evaluated under the equal-budget metaheuristic optimization framework: (a) Random Forest, (b) Support Vector Machine, (c) Multilayer Perceptron, and (d) 1D Convolutional Neural Network.

- **1D-CNN:** Filter count  $n_{\text{filter}} \in [8, 128]$ , kernel size  $k_{\text{size}} \in [2, 5]$ , dense-layer width  $n_{\text{dense}} \in [16, 256]$ , L2 penalty coefficient  $\alpha \in [10^{-8}, 10^{-2}]$ , Adam learning rate  $\eta \in [10^{-8}, 10^{-2}]$ , dropout  $p_{\text{drop}} \in [0, 0.10]$ , and batch size in  $\{16, 32, 64, 80, 112, 128\}$ .

### 3.5 Optimizer Configuration, Search Operators, and Validation Objectives

To ensure complete experimental reproducibility, five search paradigms were evaluated: Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO). All optimizers operate under a strict candidate evaluation cap of  $N_{\text{eval}} = 1,000$  per model and random seed. For all population-based algorithms (GA, PSO, DE, GWO), the population size is fixed to  $N_{\text{pop}} = 10$  candidates evaluated across  $G = 100$  generations or iterations ( $10 \times 100 = 1,000$  function evaluations). RS generates  $N_{\text{eval}} = 1,000$  independent uniform random samples across the search space.

Candidate vectors use the bounds stated above. Integer parameters and categorical-option indices are rounded and clipped to their admissible bounds:

$$\theta_d = \text{clip}(\text{round}(u_d), \theta_{d,\min}, \theta_{d,\max}). \quad (2)$$

For continuous parameters defined on logarithmic scales (such as  $C$ ,  $\alpha$ , and learning rates), optimizers act on  $u_d \in [\log_{10}(\theta_{d,\min}), \log_{10}(\theta_{d,\max})]$  and values are decoded by

$$\theta_d = 10^{u_d}. \quad (3)$$

The specific search operators and update formulations governing each optimizer are defined as follows:

- **Genetic Algorithm (GA)**: Utilizes tournament selection ( $k = 3$ ), uniform crossover with probability  $P_c = 0.80$ , and mutation with probability  $P_m = 0.20$ . When a gene is mutated, its perturbation scale is 0.10 of the corresponding encoded search range.
- **Particle Swarm Optimization (PSO)**: Velocity vectors  $\mathbf{v}_i$  and positions  $\mathbf{u}_i$  for particle  $i$  at iteration  $g$  are updated using inertia weight  $w = 0.70$  and cognitive/social coefficients  $c_1 = c_2 = 1.50$ :

$$\mathbf{v}_i^{g+1} = w\mathbf{v}_i^g + c_1 r_1 (\mathbf{pbest}_i - \mathbf{u}_i^g) + c_2 r_2 (\mathbf{gbest} - \mathbf{u}_i^g), \quad (4)$$

$$\mathbf{u}_i^{g+1} = \text{clip} \left( \mathbf{u}_i^g + \mathbf{v}_i^{g+1}, 0, 1 \right), \quad (5)$$

where  $r_1, r_2 \sim U(0, 1)$ .

- **Differential Evolution (DE)**: Implements the DE/best/1/bin strategy with mutation factor  $F = 0.80$  and binomial crossover probability  $CR = 0.90$ :

$$\mathbf{v}_i^{g+1} = \mathbf{gbest}^g + F(\mathbf{u}_{r_1}^g - \mathbf{u}_{r_2}^g), \quad (6)$$

where  $r_1, r_2$  are distinct randomly chosen population indices.

- **Grey Wolf Optimizer (GWO)**: Candidate position vectors are updated by simulating encirclement of the three leading wolves ( $\alpha, \beta, \delta$ ):

$$\mathbf{u}_i^{g+1} = \frac{\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3}{3}, \quad (7)$$

where  $\mathbf{x}_1 = \alpha - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \alpha - \mathbf{u}_i^g|$ , with analogous terms for  $\mathbf{x}_2$  ( $\beta$ ) and  $\mathbf{x}_3$  ( $\delta$ ).

Candidate configurations are evaluated against two separated validation objectives:

- **Protocol A (MCC/ $F_1$  Compound Objective)**:

$$f_{\text{MCC}/F_1}(\theta) = 0.60 \text{MCC}_{\text{val}}(\theta) + 0.40 F_{1,\text{val}}(\theta). \quad (8)$$

- **Protocol B (Weighted Accuracy Objective)**:

$$f_{\text{Acc}}(\theta) = 0.40 \text{Accuracy}_{\text{train}}(\theta) + 0.60 \text{Accuracy}_{\text{val}}(\theta). \quad (9)$$

Upon completion of the  $N_{\text{eval}} = 1,000$  search budget, the optimal candidate vector  $\theta^* = \arg \max f(\theta)$  is refit on the concatenated training and validation dataset ( $N = 12,045$  bars) and evaluated exactly once on the locked test block ( $N = 3,012$  bars). The exploratory economic assessment maps an uptrend prediction to a long position and a downtrend prediction to a short position, opened at the bar open and closed at its close. Each bar is treated as one round-trip trade and is charged a fixed cost of five index points. We report total net profit, maximum drawdown, profit factor, and annualized Sharpe ratio; these quantities are therefore specific to this simplified intrabar execution rule rather than a claim of deployable live performance.

### 3.6 Statistical Analysis

Statistical comparisons are conducted separately within each holdout protocol and use the four classifier backbones as the groups of interest. For a given endpoint, each optimizer–seed pair defines one matched block, giving 15 blocks (five optimizers and three seeds) per protocol. We first apply the Friedman rank test to assess whether model outcomes differ across these matched blocks. When summarizing pairwise contrasts, the model with the highest marginal mean for the endpoint is compared with each of the other three models using two-sided paired Wilcoxon signed-rank tests. The resulting three  $p$ -values are adjusted together using the Holm step-down procedure, with family-wise significance level  $\alpha = 0.05$ .

These tests are exploratory. The blocks share one chronological test segment, and the five optimizers and three seeds do not provide 15 independent market periods. Accordingly, the tests quantify consistency of model ordering across the evaluated optimizer–seed combinations on this fixed test period; they do not establish replication across market regimes or a confirmatory difference between validation protocols. No optimizer-superiority claim is inferred from these model-level tests.

## 4 Results

This section presents the experimental results and their implications for short-term prediction of Brazilian Mini-Index futures. Following the organization of the precursor, it first describes the common experimental setup and then discusses the performance of the individual model families. The expanded benchmark subsequently reports the direct cross-model comparison, exploratory statistical evidence, and economic outcomes.

### 4.1 Experimental Setup and Data Partitioning

All model–optimizer combinations use the seven descriptors described in Section 3.3, the chronological split shown in Figure 2, and the same 1,000-evaluation allowance per seed. RF, SVM, MLP, and 1D-CNN each are searched with RS, GA, PSO, DE, and GWO. The two protocols differ only in the validation fitness used to rank candidate configurations. Three common seeds (1–3) are available for every cell in both protocols.

Consequently, the direct comparison between fitness functions is descriptive: each reported model–optimizer result is the mean over the same three seeds. The locked test block contains 3,012 observations and is not used during search. This design differs from the randomized training and test sampling used in the precursor and removes look-ahead exposure from model selection.

### 4.2 Reference Baselines

Two deterministic reference models were evaluated on the same locked test block before interpreting the optimizer benchmark. The majority-direction baseline predicts the class that is more frequent in the training partition; its probability is fixed at that training prevalence. The default RF baseline uses the same fixed seven-feature representation and the final-refit rule (training plus validation), but no hyperparameter

search. These references are not additional optimizers and do not enter the matched optimizer–seed statistical tests.

Table 3 illustrates why accuracy or positive-class  $F_1$  alone are insufficient in this setting. The naive baseline predicts every test observation as an uptrend, obtaining  $F_1 = 0.675$  because the positive class is common, but MCC is exactly zero. The default RF yields MCC 0.210 and balanced accuracy 0.604, establishing a temporal no-tuning reference. The optimized model summaries reported below must therefore be read as evidence beyond both a trivial directional rule and this default classifier, while acknowledging that the default RF is only one backbone-specific reference.

**Table 3** Locked-test reference baselines under the temporal protocol. The naive baseline selects the training-majority direction (uptrend); the default RF is refitted on training plus validation with 100 trees and no hyperparameter search.

| Reference                | Accuracy | Balanced accuracy | $F_1$ | MCC   | AUC-ROC |
|--------------------------|----------|-------------------|-------|-------|---------|
| Naive majority direction | 0.510    | 0.500             | 0.675 | 0.000 | 0.500   |
| Default RF (no tuning)   | 0.605    | 0.604             | 0.630 | 0.210 | 0.645   |

### 4.3 Performance of the Models

Table 4 retains the presentation pattern of the precursor: it states the search space and an illustrative GA-selected configuration for each of the four model families. The configurations are from seed 1 under Protocol A and are included to make the optimization procedure concrete; they are not asserted to be universally optimal because all optimizer–seed outcomes are summarized below.

Figures 4–7 replace the three GA-only convergence plots of the precursor with the five budget-matched search trajectories. For each model, the left panel follows the compound MCC/ $F_1$  fitness and the right panel follows weighted training/validation accuracy. The vertical scales differ because the objectives differ; their role is to show search convergence within a protocol, not to compare objective magnitudes across protocols.

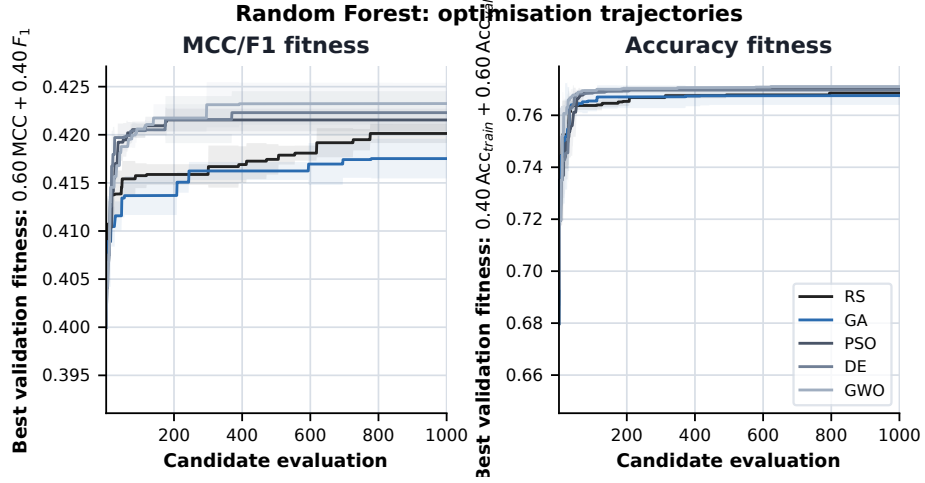
### 4.4 Optimizer-Level Performance Under Equal Budget

Table 5 exposes the comparison that is obscured when results are first aggregated across optimizers. Every cell is the mean locked-test primary metric across the same three seeds, and every optimizer consumed the same 1,000 candidate evaluations. The table is descriptive rather than inferential: it establishes the observed optimizer-by-backbone pattern, while the small number of seeds prevents a reliable universal ranking of optimizers.

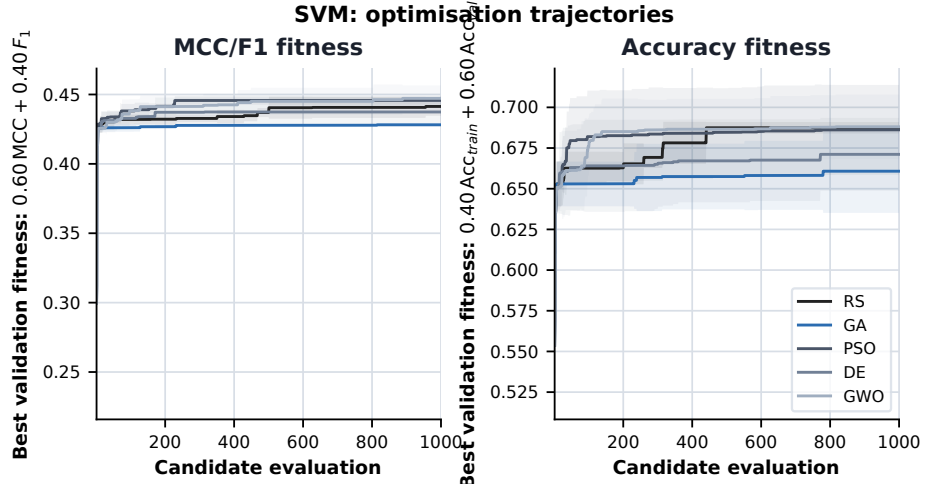
Under MCC/ $F_1$  fitness, GA, PSO, and DE reach similarly high mean MCC values for RF, whereas the MLP rows favour the population methods over Random Search. The pattern is less uniform for SVM and 1D-CNN, for which the ordering changes with the optimizer. Under weighted-accuracy fitness, MLP remains close to 0.65 accuracy for all five methods, while RF and SVM are more sensitive to the optimizer choice.

**Table 4** Hyperparameter search domains and representative GA-selected configurations across the four evaluated classifier backbones.

| Random Forest (RF)          |                                  |                       |                      | Support Vector Machine (SVM)             |                       |                |              |
|-----------------------------|----------------------------------|-----------------------|----------------------|------------------------------------------|-----------------------|----------------|--------------|
| Hyperparameter              | Search space                     | Selected              | Hyperparameter       | Search space                             | Selected              | Hyperparameter | Search space |
| $n_{\text{estimators}}$     | [50, 500]                        | 58                    | $C$                  | $[10^{-3}, 10^3]$ (log scale)            | 2.089                 |                |              |
| max_depth                   | [5, 50]                          | 6                     | $\gamma$             | $[10^{-5}, 10^1]$ (log scale)            | 5.044                 |                |              |
| min_samples_split           | [2, 20]                          | 13                    | Kernel               | {linear, RBF}                            | linear                |                |              |
| min_samples_leaf            | [1, 20]                          | 15                    |                      |                                          |                       |                |              |
| max_features                | {sqrt, log2, None}               | None                  |                      |                                          |                       |                |              |
| class_weight                | {None, balanced}                 | None                  |                      |                                          |                       |                |              |
| Multilayer Perceptron (MLP) |                                  |                       |                      | 1D Convolutional Neural Network (1D-CNN) |                       |                |              |
| Hyperparameter              | Search space                     | Selected              | Hyperparameter       | Search space                             | Selected              | Hyperparameter | Search space |
| Hidden neurons              | [5, 500]                         | 410                   | Filters              | [8, 128]                                 | 48                    |                |              |
| Learning rate               | $[10^{-8}, 10^{-2}]$ (log scale) | $3.42 \times 10^{-4}$ | Kernel size          | [2, 5]                                   | 3                     |                |              |
| $L_2$ regularization        | $[10^{-8}, 10^{-2}]$ (log scale) | $1.66 \times 10^{-7}$ | Dense neurons        | [16, 256]                                | 251                   |                |              |
| Dropout rate                | [0, 0.10]                        | 0.057                 | Learning rate        | $[10^{-8}, 10^{-2}]$ (log scale)         | $3.90 \times 10^{-6}$ |                |              |
| Batch size                  | {16, 32, 64, 80, 112, 128}       | 112                   | $L_2$ regularization | $[10^{-8}, 10^{-2}]$ (log scale)         | $5.70 \times 10^{-6}$ |                |              |
|                             |                                  |                       | Dropout rate         | [0, 0.10]                                | 0.026                 |                |              |
|                             |                                  |                       | Batch size           | {16, 32, 64, 80, 112, 128}               | 16                    |                |              |



**Fig. 4** Fitness evolution during equal-budget optimization of the Random Forest model. Lines summarize the three common seeds.

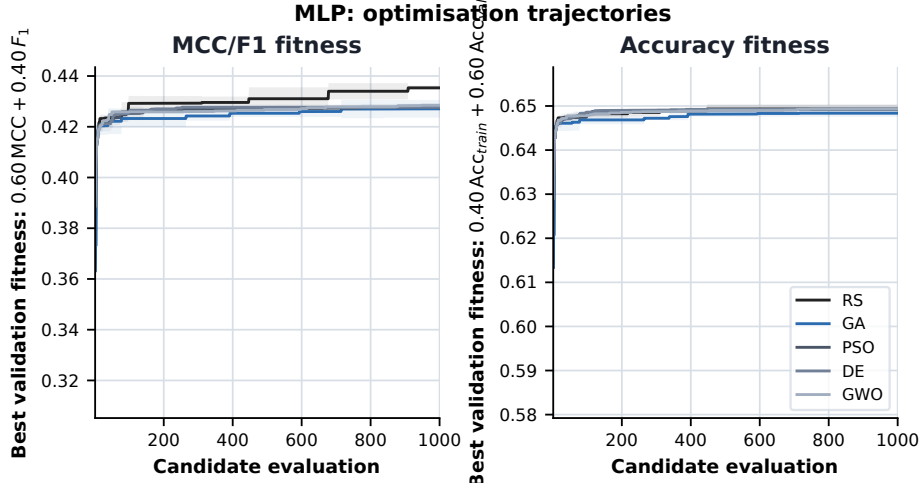


**Fig. 5** Fitness evolution during equal-budget optimization of the SVM model. Lines summarize the three common seeds.

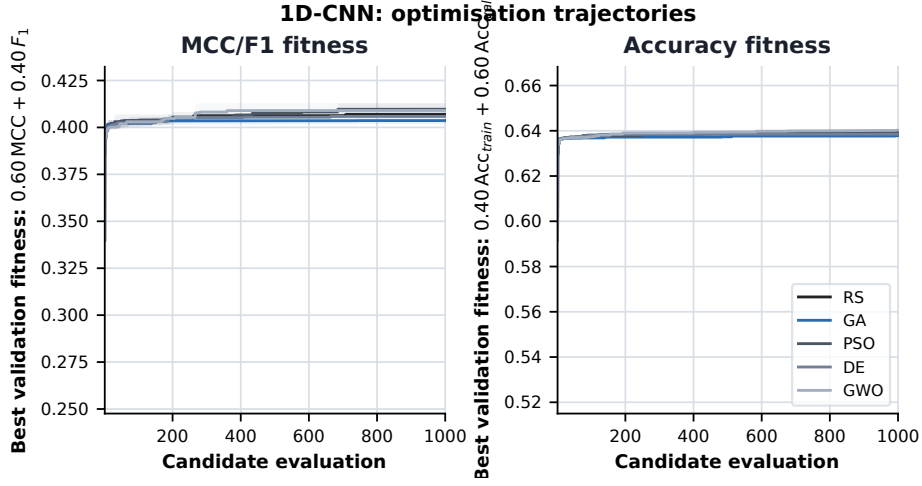
These contrasts motivate the model-level aggregation in the next subsection, but they also show why an optimizer cannot be declared superior from one pooled summary.

#### 4.5 Model-Level Aggregation Across Optimizers

Table 6 and Figure 8 present locked-test performance aggregated over 15 optimizer-seed blocks per model within each protocol. Reporting the standard deviation is essential here: the blocks combine five optimizers and three common seeds, so the



**Fig. 6** Fitness evolution during equal-budget optimization of the MLP model. Lines summarize the three common seeds.



**Fig. 7** Fitness evolution during equal-budget optimization of the 1D-CNN model. Lines summarize the three common seeds.

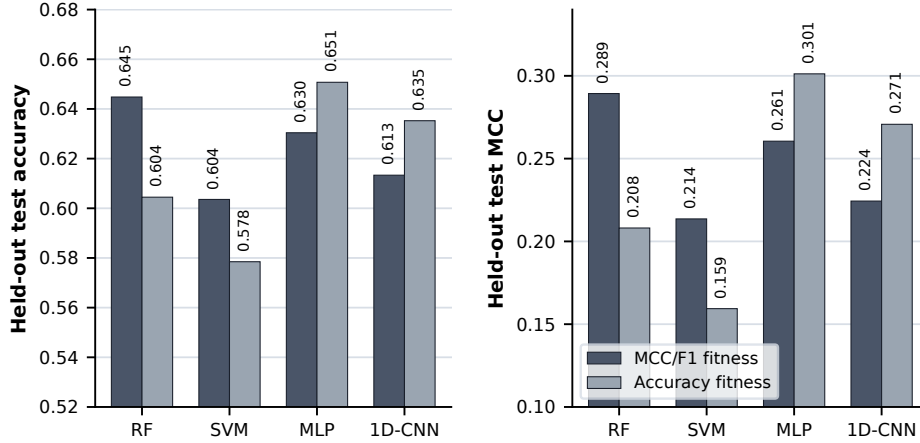
dispersion describes sensitivity to the evaluated search procedure rather than independent market replications. Under MCC/ $F_1$  fitness, RF has the largest mean test accuracy (0.645), MCC (0.289), and  $F_1$  (0.664), with markedly lower variability than the other backbones. Under weighted-accuracy fitness, MLP has the largest mean accuracy (0.651), MCC (0.301), and  $F_1$  (0.658). Thus, the validation objective changes the descriptive ordering of the model families and should not be treated as interchangeable with the final reporting metric.

**Table 5** Optimizer-level locked-test primary metrics under equal budget. Each cell is a mean over three common seeds; boldface denotes the largest mean within a model and protocol. RS: Random Search; GA: Genetic Algorithm; PSO: Particle Swarm Optimization; DE: Differential Evolution; GWO: Grey Wolf Optimizer.

| Model                                                           | RS    | GA           | PSO          | DE    | GWO          |
|-----------------------------------------------------------------|-------|--------------|--------------|-------|--------------|
| <b>Panel A: MCC/<math>F_1</math> fitness; locked-test MCC</b>   |       |              |              |       |              |
| RF                                                              | 0.281 | <b>0.293</b> | 0.293        | 0.292 | 0.288        |
| SVM                                                             | 0.105 | <b>0.290</b> | 0.255        | 0.187 | 0.232        |
| MLP                                                             | 0.106 | 0.290        | 0.299        | 0.302 | <b>0.305</b> |
| 1D-CNN                                                          | 0.179 | 0.268        | <b>0.269</b> | 0.267 | 0.139        |
| <b>Panel B: weighted-accuracy fitness; locked-test accuracy</b> |       |              |              |       |              |
| RF                                                              | 0.604 | <b>0.611</b> | 0.604        | 0.600 | 0.602        |
| SVM                                                             | 0.588 | 0.557        | 0.571        | 0.584 | <b>0.593</b> |
| MLP                                                             | 0.649 | 0.649        | <b>0.652</b> | 0.652 | 0.652        |
| 1D-CNN                                                          | 0.636 | 0.634        | 0.635        | 0.634 | <b>0.637</b> |

**Table 6** Locked-test predictive outcomes by backbone and validation protocol. Entries are mean  $\pm$  standard deviation over 15 optimizer-seed blocks per row (five optimizers  $\times$  three common seeds).

| Model  | MCC/ $F_1$ fitness |                   |                   | Weighted accuracy fitness |                   |                   |
|--------|--------------------|-------------------|-------------------|---------------------------|-------------------|-------------------|
|        | Accuracy           | MCC               | $F_1$             | Accuracy                  | MCC               | $F_1$             |
| RF     | $0.645 \pm 0.004$  | $0.289 \pm 0.008$ | $0.664 \pm 0.006$ | $0.604 \pm 0.005$         | $0.208 \pm 0.009$ | $0.624 \pm 0.004$ |
| SVM    | $0.604 \pm 0.060$  | $0.214 \pm 0.132$ | $0.602 \pm 0.122$ | $0.578 \pm 0.028$         | $0.159 \pm 0.053$ | $0.600 \pm 0.071$ |
| MLP    | $0.630 \pm 0.070$  | $0.261 \pm 0.141$ | $0.639 \pm 0.071$ | $0.651 \pm 0.003$         | $0.301 \pm 0.005$ | $0.658 \pm 0.004$ |
| 1D-CNN | $0.613 \pm 0.056$  | $0.224 \pm 0.120$ | $0.597 \pm 0.166$ | $0.635 \pm 0.002$         | $0.271 \pm 0.005$ | $0.639 \pm 0.007$ |



**Fig. 8** Comparison of mean locked-test accuracy and MCC for all four model families under the two validation objectives.

## 4.6 Exploratory Statistical Evidence

The model-level tests are performed separately within each holdout protocol. The five optimizer–seed combinations yield 15 matched blocks for each backbone. Table 7 reports the two preselected endpoint tests that correspond to the respective validation objectives. Under MCC/ $F_1$  fitness, the Friedman comparison of locked-test MCC is significant ( $\chi^2 = 22.84$ ,  $p = 4.36 \times 10^{-5}$ ). RF has the largest marginal mean MCC (Table 6), whereas MLP has the best mean rank across the matched blocks; this distinction is precisely why both summaries are reported. Under weighted-accuracy fitness, the accuracy comparison is significant ( $\chi^2 = 42.92$ ,  $p = 2.56 \times 10^{-9}$ ), and MLP is first both by mean accuracy and mean rank. Its Holm-adjusted contrasts against RF, SVM, and 1D-CNN are all significant ( $p_{\text{Holm}} \approx 0.00194$ ).

**Table 7** Exploratory Friedman tests for the endpoint aligned with each validation protocol. Lower mean rank is better. The blocks are optimizer–seed combinations from one locked chronological segment and are not independent market replications.

| Protocol                  | Endpoint      | $n$ | $\chi^2$ | $p$                   | RF   | SVM  | MLP  | 1D-CNN |
|---------------------------|---------------|-----|----------|-----------------------|------|------|------|--------|
| Mean rank                 |               |     |          |                       |      |      |      |        |
| MCC/ $F_1$ fitness        | Test MCC      | 15  | 22.84    | $4.36 \times 10^{-5}$ | 2.07 | 2.73 | 1.53 | 3.67   |
| Weighted accuracy fitness | Test accuracy | 15  | 42.92    | $2.56 \times 10^{-9}$ | 3.13 | 3.87 | 1.00 | 2.00   |

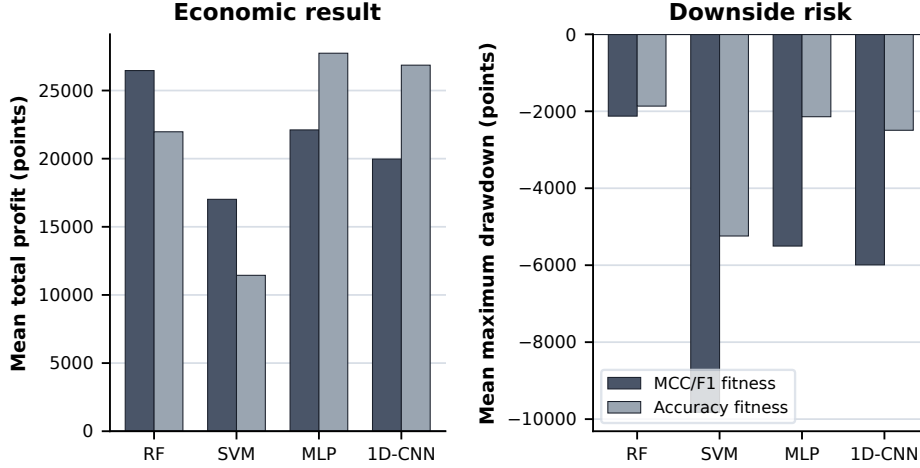
Because these blocks share one chronological test segment and only three random seeds, the tests summarize variation across the evaluated optimizer–seed combinations. They do not establish independent replication across market periods, nor do they test a confirmatory difference between the two fitness functions. The inferential results should therefore be read together with the descriptive dispersion in Table 6, rather than as a claim that one backbone is universally superior.

## 4.7 Exploratory Economic Evaluation

The economic evaluation yields a complementary ranking under the fixed intrabar execution rule. Table 8 reports the complete set of economic endpoints rather than total profit alone. Under MCC/ $F_1$  fitness, RF has the largest mean total profit (26,465 points), the highest mean profit factor, and the smallest drawdown dispersion. Under weighted-accuracy fitness, MLP has the largest mean total profit (27,737 points) and profit factor, whereas RF has the least negative mean maximum drawdown (-1,866 points). Thus, the predictive winner and the return–risk profile do not always coincide. Figure 9 visualizes the profit–drawdown component of this comparison; it should not be read as a transaction-cost-complete trading recommendation.

**Table 8** Exploratory economic outcomes under the fixed intrabar execution rule. Net profit and maximum drawdown (Max. DD) are points; PF denotes profit factor and Sharpe is annualized. Entries are mean  $\pm$  standard deviation over 15 optimizer–seed blocks.

| Protocol                  | Model  | Net profit          | Max. DD             | PF   | Sharpe |
|---------------------------|--------|---------------------|---------------------|------|--------|
| MCC/ $F_1$ fitness        | RF     | 26,465 $\pm$ 1,732  | −2,126 $\pm$ 141    | 1.48 | 12.50  |
|                           | SVM    | 17,017 $\pm$ 29,437 | −9,823 $\pm$ 17,123 | 1.39 | 8.52   |
|                           | MLP    | 22,110 $\pm$ 20,556 | −5,503 $\pm$ 12,984 | 1.43 | 10.62  |
|                           | 1D-CNN | 19,969 $\pm$ 17,574 | −5,991 $\pm$ 10,469 | 1.37 | 7.67   |
| Weighted accuracy fitness | RF     | 21,973 $\pm$ 1,336  | −1,866 $\pm$ 365    | 1.38 | 11.95  |
|                           | SVM    | 11,440 $\pm$ 13,350 | −5,241 $\pm$ 1,961  | 1.21 | 4.18   |
|                           | MLP    | 27,737 $\pm$ 1,069  | −2,141 $\pm$ 112    | 1.51 | 13.07  |
|                           | 1D-CNN | 26,861 $\pm$ 1,924  | −2,494 $\pm$ 62     | 1.49 | 9.71   |



**Fig. 9** Mean total profit and maximum drawdown across 15 optimizer–seed blocks per model. Results are reported separately for the two validation objectives.

## 5 Discussion

The purpose of this benchmark is not to replace the practical pipeline of the blinded precursor, but to determine which conclusions remain when the temporal split, optimizer budget, validation objective, and economic endpoint are made explicit. The resulting evidence separates a model’s average locked-test outcome, its sensitivity to the optimizer–seed block, and its simplified trading consequence. This distinction matters for intraday financial prediction, where a change in selection criterion can alter a model ranking without demonstrating a stable change in market-wide forecasting ability.

### 5.1 What the Controlled Comparison Establishes

The present design preserves the precursor’s technical-indicator workflow and fixed seven-feature representation, but changes the comparison unit. Rather than reporting

a GA-selected configuration after randomized cross-validation, every backbone is evaluated with RS, GA, PSO, DE, and GWO under the same 1,000-candidate allowance and the same chronological train/validation/test partition. Random Search is therefore a budget-matched reference against which directed population methods can be interpreted [10]. The common test segment also prevents final performance values from selecting hyperparameters.

This control improves internal comparability, but it narrows the claim that the data can support. The evidence concerns four classifier families on one five-minute WIN period, conditional on an inherited input representation. It does not show that the seven indicators would be selected again in another training period, nor that an optimizer with a favourable result here will retain that advantage under another market regime. The contribution is consequently a controlled experimental frame, not a universal ordering of metaheuristics.

## 5.2 Predictive Performance Depends on the Validation Objective

The locked-test summaries in Table 6 show a clear reversal at the model level. Under the compound  $MCC/F_1$  objective, RF has the largest marginal means for accuracy, MCC, and  $F_1$ , and its small standard deviations indicate comparatively low sensitivity to the evaluated optimizer–seed blocks. Under weighted validation accuracy, MLP is first on all three reported predictive measures, while 1D-CNN is also stable but remains below MLP in mean accuracy and MCC. The change in ordering is substantive: selecting for validation accuracy does not simply rescale the  $MCC/F_1$  result.

MCC is useful because it summarizes the association between predicted and observed binary classes beyond raw accuracy [3]; however, optimizing an imbalance-aware criterion cannot guarantee the best outcome for every backbone and search trajectory. Conversely, the MLP advantage under weighted accuracy is concentrated in a low-variance regime, whereas MLP, SVM, and 1D-CNN outcomes under  $MCC/F_1$  fitness vary much more across the evaluated blocks. A reported mean must therefore be accompanied by its dispersion and by the selection rule that generated it.

The rank analysis reinforces this caution. In the  $MCC/F_1$  protocol, RF has the highest marginal mean MCC, but MLP has the lowest mean Friedman rank (Table 7). This is not a contradiction: means weight score magnitude, whereas ranks summarize ordering within each matched optimizer–seed block. Under weighted accuracy, both summaries place MLP first, and its Holm-adjusted contrasts against all other backbones are significant. Because all 15 blocks share one locked chronological segment, these results demonstrate separation within the evaluated design rather than independent replication across time.

## 5.3 Optimizer Trajectories Are Diagnostic, Not a Universal League Table

Figures 4–7 are valuable because every trajectory is indexed by candidate evaluations rather than by a method-specific generation count. Initial gains reveal how quickly a

method locates workable regions of a search space, whereas later plateaus reveal the incremental benefit of further evaluations under the fixed budget. In several panels, methods converge to close validation values despite different update mechanisms. Such convergence does not itself establish equivalent generalization: the locked-test values and dispersion in Table 6 remain necessary for that interpretation.

Search behaviour interacts with the backbone and the objective. A method that improves early for RF need not be the most reliable procedure for MLP or 1D-CNN, whose mixed discrete–continuous hyperparameter spaces have different response surfaces. The article therefore does not claim a universal winner among RS, GA, PSO, DE, and GWO. A stronger optimizer-level claim would require a dedicated analysis across more seeds and temporal replications, with the optimizer itself as the inferential factor.

## 5.4 Predictive Quality and Economic Outcome Are Distinct Endpoints

Table 8 makes the distinction between predictive evaluation and the exploratory trading rule concrete. Under  $MCC/F_1$  fitness, RF combines the largest mean net profit with the smallest average drawdown and the lowest drawdown dispersion. Under weighted accuracy, MLP has the largest mean net profit, profit factor, and annualized Sharpe ratio, while RF retains the least negative average drawdown. Thus, the preferred choice depends on whether the decision criterion prioritizes return, drawdown, or consistency; it cannot be inferred from accuracy alone.

These values are deliberately reported as points under a fixed intrabar long/short rule with a five-point round-trip cost. They are not an investable-strategy claim: the evaluation does not model a full transaction-cost schedule, market impact, delayed execution, position sizing, or a live signal-selection policy. Its role is diagnostic: it shows that converting a directional label to trades can change the practical ranking of model families, and that economic reporting should accompany predictive metrics.

## 5.5 Limitations and Research Agenda

Four limitations delimit the present evidence. First, only three seeds are common to every model–optimizer cell; standard deviations and rank tests describe the evaluated blocks, not repeated market samples. Second, the locked test covers one instrument and one chronological segment, so regime sensitivity and cross-market transfer remain unknown. Third, the Information-Gain mask was inherited from the blinded precursor rather than re-estimated on the current training segment; all conclusions are conditional on that representation. Fourth, equalizing the budget at 1,000 evaluations establishes fairness at one computational scale but does not characterize larger-budget behaviour or wall-clock efficiency.

The next study should preserve the current leakage-aware split and add a pre-registered common 20-seed design across multiple non-overlapping market periods. It should extend the present naive and default-RF references to no-tuning baselines for every backbone; archive every candidate evaluation; report compute time; and test

optimizer differences separately from model differences. Re-estimating feature selection within each training window would further distinguish the effect of representation from hyperparameter search. These additions would turn the current transparent benchmark into confirmatory evidence without weakening its temporal protocol.

## 6 Conclusion

This study transforms a GA-based conference precursor into a controlled Neural Computing and Applications benchmark for neural intraday trend classification. It preserves the predecessor’s practical insight—a compact, Information-Gain-selected technical representation can support metaheuristic hyperparameter search—but replaces randomized validation and a single-optimizer design with a chronological dual-fitness protocol, five budget-matched optimizers, and locked predictive and economic assessment.

The current evidence indicates that the choice between compound  $MCC/F_1$  fitness and accuracy fitness produces configuration-dependent changes in both held-out accuracy and MCC. Within the accuracy-fitness protocol, MLP is the strongest model in the exploratory predictive analysis and leads mean total profit, although RF has the smallest mean drawdown. These results reinforce the central methodological lesson: selection objective, reported metric, and economic utility must be treated as related but distinct quantities.

The manuscript deliberately avoids a confirmatory claim about a universal best fitness function because only three paired seeds are currently available across every cell. Completing the common 20-seed design is the necessary next step. The resulting framework nevertheless provides a transparent and reproducible basis for future optimizer comparisons in non-stationary financial time series.

## Declarations

### Funding

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### Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

### Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

## Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

## Author Contributions

Author-contribution information is withheld during double-blind review and will be supplied in the title page after editorial decision.

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